

Polyak-Lojasiewicz Stochastic Gradient Descent (PLSGD)

1 Algorithm Description

The algorithm iteratively updates parameters by computing gradients on randomly sampled data points and taking steps in the negative gradient direction with a fixed step size. It is designed for smooth functions satisfying the Polyak-Lojasiewicz condition.

1.1 Mathematical Formulation

Objective Function: Minimize a smooth function $f(w) : \mathbb{R}^d \rightarrow \mathbb{R}$, where $f(w) = \mathbb{E}_z[f(w; z)]$, and z represents a random data point.

Assumptions:

1. **Smoothness:** f is L -smooth, i.e., $\|\nabla f(w) - \nabla f(w')\| \leq L\|w - w'\|$ for all w, w' .
2. **Polyak-Lojasiewicz (PL) Condition:** There exists $\mu > 0$ such that:

$$\|\nabla f(w)\|^2 \geq 2\mu(f(w) - f^*) \quad \forall w,$$

where $f^* = \min_w f(w)$ is the global minimum.

Update Rule: At iteration t , sample a mini-batch B_t of size B , compute the gradient estimate:

$$g_t = \frac{1}{B} \sum_{z \in B_t} \nabla f(w_t; z),$$

and update the parameters via:

$$w_{t+1} = w_t - \eta g_t,$$

where $\eta > 0$ is a fixed step size.

Hyperparameters:

- Step size η : Controls the magnitude of parameter updates.
- Mini-batch size B : Balances gradient estimation noise and computational cost.
- Maximum iterations T : Upper bound on algorithm runtime.
- Tolerance ε : Termination threshold for gradient norm (optional).

2 Pseudo Code and Dry Run

2.1 Pseudocode

Input: Step size $\eta > 0$, mini-batch size B , maximum iterations T , tolerance $\varepsilon \geq 0$. **Initialize:** Parameter vector $w_0 \in \mathbb{R}^d$. $t = 0$ to $T - 1$ Sample mini-batch B_t of B data points. Compute gradient estimate: $g_t = \frac{1}{B} \sum_{z \in B_t} \nabla f(w_t; z)$. Update parameters: $w_{t+1} = w_t - \eta g_t$. $\|g_t\| < \varepsilon$ Terminate early. **Output:** Final parameter vector w_T .

2.2 Dry Run Example

Function: $f(w) = (w - 3)^2$. **Initialization:** $w_0 = 0$, $\eta = 0.1$, mini-batch size $B = 1$.

Iteration 1:

- Gradient: $\nabla f(w_0) = 2(w_0 - 3) = -6$.
- Update: $w_1 = 0 - 0.1 \cdot (-6) = 0.6$.
- Objective: $f(0.6) = (0.6 - 3)^2 = 5.76$.

Iteration 2:

- Gradient: $\nabla f(w_1) = 2(0.6 - 3) = -4.8$.
- Update: $w_2 = 0.6 - 0.1 \cdot (-4.8) = 1.08$.
- Objective: $f(1.08) = (1.08 - 3)^2 = 3.6864$.

Iteration 3:

- Gradient: $\nabla f(w_2) = 2(1.08 - 3) = -3.84$.
- Update: $w_3 = 1.08 - 0.1 \cdot (-3.84) = 1.464$.
- Objective: $f(1.464) = (1.464 - 3)^2 = 2.359296$.

Observation: The iterates converge monotonically toward the minimum at $w = 3$, demonstrating geometric convergence consistent with the PL condition.

3 Convergence Theory

3.1 Assumptions

We formalize the assumptions required for our convergence analysis:

[(A)]**Smoothness.** The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable and L -smooth. That is, its gradient is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

This guarantees that the function does not exhibit curvature sharper than L . **Polyak-Lojasiewicz (PL) Condition.** The function f satisfies the PL inequality with constant $\mu > 0$:

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x \in \mathbb{R}^n,$$

where f^* is the global minimum of f . This condition ensures that the function's gradient norm bounds its suboptimality, even when f is non-convex. **Parameter Constraints.** The fixed step size η satisfies $0 < \eta \leq \frac{1}{L}$. This ensures stability of gradient descent updates.

3.2 Main Convergence Theorem

We now state and prove the convergence rate of gradient descent under assumptions (A)–(C).

[Linear Convergence] Let f satisfy (A)–(C). Gradient descent with step size $\eta = \frac{1}{L}$ satisfies:

$$f(x_T) - f^* \leq \left(1 - \frac{\mu}{L}\right)^T (f(x_0) - f^*).$$

Thus, the iterates converge linearly to the global minimum f^* .

By L -smoothness, the gradient descent update $x_{t+1} = x_t - \eta \nabla f(x_t)$ satisfies:

$$f(x_{t+1}) \leq f(x_t) - \eta \|\nabla f(x_t)\|^2 + \frac{L\eta^2}{2} \|\nabla f(x_t)\|^2.$$

Substituting $\eta = \frac{1}{L}$:

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L} \|\nabla f(x_t)\|^2.$$

Using the PL inequality $\|\nabla f(x_t)\|^2 \geq 2\mu(f(x_t) - f^*)$, we obtain:

$$f(x_{t+1}) - f^* \leq \left(1 - \frac{\mu}{L}\right) (f(x_t) - f^*).$$

Unrolling this recursion completes the proof.

Special Cases:

- 3. If f is μ -strongly convex, it satisfies the PL condition with constant μ , and this result recovers the classical linear rate of gradient descent.
- When $\mu = L$, the rate becomes exponential: $f(x_T) - f^* \leq e^{-T}(f(x_0) - f^*)$.

3.3 Theoretical Properties

3.3.1 Stability Analysis

The step size $\eta \leq \frac{1}{L}$ ensures that the update does not overshoot the curvature of f . If $\eta > \frac{2}{L}$, gradient descent may diverge even for convex f .

3.3.2 Sensitivity to Hyperparameters

The convergence rate depends critically on μ and L . A smaller μ/L ratio implies slower convergence. Estimating L accurately is essential for setting η ; overestimating L (and thus underestimating η) reduces the effective rate.

3.3.3 Comparison to Baseline Methods

- **Strongly Convex Functions:** Gradient descent on μ -strongly convex functions achieves the same linear rate $(1 - \mu/L)$, but the PL condition allows extension to non-convex functions.
- **Convex Functions:** For convex f without the PL condition, gradient descent only guarantees sub-linear convergence $O(1/T)$.

3.4 Discussion

This analysis shows that gradient descent with fixed step size exhibits linear convergence for smooth functions satisfying the PL condition—a significant improvement over sublinear rates for general convex functions. The PL condition, while less restrictive than strong convexity, ensures that all critical points are global minima and provides a direct link between gradient norm and suboptimality.

The algorithm described in the problem—using gradients of randomly sampled data points—resembles stochastic gradient descent (SGD). However, our analysis assumes exact gradients. For SGD extensions, an additional bounded variance assumption on stochastic gradients is typically required to establish convergence to a neighborhood of f^* . In this case, the convergence rate remains linear up to a noise floor dependent on the gradient variance.

Future work could extend these results to the stochastic setting, analyze adaptive step size strategies, or explore connections to distributed optimization under the PL condition.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

1. **L-Smoothness:** A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth if it is twice differentiable and its gradient is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

This implies that the Hessian $\nabla^2 f(x)$ satisfies $\nabla^2 f(x) \preceq LI$ for all x .

2. **Polyak-Lojasiewicz (PL) Condition:** A function f satisfies the PL inequality with constant $\mu > 0$ if:

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x \in \mathbb{R}^n,$$

where f^* is the global minimum of f . This condition ensures that the gradient norm bounds the suboptimality of $f(x)$.

3. **Step Size Constraint:** The fixed step size η satisfies $0 < \eta \leq \frac{1}{L}$, ensuring stability of gradient descent updates.

4.1.2 Lemmas

Lemma 1 (Descent Lemma): For an L -smooth function f , the following inequality holds for all $x, y \in \mathbb{R}^n$:

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2}\|y - x\|^2.$$

Proof: By the fundamental theorem of calculus and the Lipschitz continuity of ∇f , we have:

$$f(y) = f(x) + \int_0^1 \nabla f(x + t(y - x))^T(y - x) dt.$$

Subtract $\nabla f(x)^T(y - x)$ from both sides:

$$f(y) - f(x) - \nabla f(x)^T(y - x) = \int_0^1 [\nabla f(x + t(y - x)) - \nabla f(x)]^T(y - x) dt.$$

Using the Cauchy-Schwarz inequality and Lipschitz continuity:

$$\left| [\nabla f(x + t(y - x)) - \nabla f(x)]^T(y - x) \right| \leq \|\nabla f(x + t(y - x)) - \nabla f(x)\| \cdot \|y - x\| \leq Lt\|y - x\|^2.$$

Integrating over $t \in [0, 1]$:

$$f(y) - f(x) - \nabla f(x)^T(y - x) \leq \int_0^1 Lt\|y - x\|^2 dt = \frac{L}{2}\|y - x\|^2.$$

Rearranging completes the proof. $\square$

Lemma 2 (PL Gradient Bound): If f satisfies the PL condition with constant μ , then for all $x \in \mathbb{R}^n$:

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*).$$

This is the definition of the PL condition and requires no further proof.

4.2 Main Proof (Step-by-Step Derivation)

We analyze the gradient descent update:

$$x_{t+1} = x_t - \eta \nabla f(x_t),$$

where $\eta = \frac{1}{L}$.

4.2.1 Step 1: Apply the Descent Lemma

Using Lemma 1 with $y = x_{t+1}$ and $x = x_t$:

$$f(x_{t+1}) \leq f(x_t) + \nabla f(x_t)^T (x_{t+1} - x_t) + \frac{L}{2} \|x_{t+1} - x_t\|^2.$$

Substitute $x_{t+1} - x_t = -\eta \nabla f(x_t)$:

$$f(x_{t+1}) \leq f(x_t) - \eta \|\nabla f(x_t)\|^2 + \frac{L\eta^2}{2} \|\nabla f(x_t)\|^2.$$

4.2.2 Step 2: Substitute $\eta = \frac{1}{L}$

Set $\eta = \frac{1}{L}$:

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{L} \|\nabla f(x_t)\|^2 + \frac{L}{2} \cdot \frac{1}{L^2} \|\nabla f(x_t)\|^2.$$

Simplify the terms:

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{L} \|\nabla f(x_t)\|^2 + \frac{1}{2L} \|\nabla f(x_t)\|^2 = f(x_t) - \frac{1}{2L} \|\nabla f(x_t)\|^2.$$

4.2.3 Step 3: Apply the PL Condition

Using Lemma 2:

$$\|\nabla f(x_t)\|^2 \geq 2\mu(f(x_t) - f^*).$$

Substitute into the inequality:

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L} \cdot 2\mu(f(x_t) - f^*) = f(x_t) - \frac{\mu}{L} (f(x_t) - f^*).$$

Subtract f^* from both sides:

$$f(x_{t+1}) - f^* \leq \left(1 - \frac{\mu}{L}\right) (f(x_t) - f^*).$$

4.2.4 Step 4: Unroll the Recursion

Iterating the inequality from $t = 0$ to $T - 1$:

$$f(x_T) - f^* \leq \left(1 - \frac{\mu}{L}\right)^T (f(x_0) - f^*).$$

This establishes linear convergence with rate $\left(1 - \frac{\mu}{L}\right)^T$. $\square$

4.3 Rate Derivation (With Constants)

The convergence rate is governed by the factor $\rho = 1 - \frac{\mu}{L}$. Since $0 < \mu \leq L$, we have $0 < \rho < 1$. Using the inequality $1 - x \leq e^{-x}$ for $x \in (0, 1)$, we bound:

$$\rho^T = \left(1 - \frac{\mu}{L}\right)^T \leq e^{-\mu T/L}.$$

Thus, the convergence rate is exponential in T , with the exponent depending on the ratio μ/L .

4.4 Special Cases

4.4.1 Case 1: Strongly Convex Functions

If f is μ -strongly convex, then for all x, y :

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{\mu}{2}\|y - x\|^2.$$

Minimizing both sides over y gives:

$$f^* \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2 \implies \|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*).$$

Thus, strong convexity implies the PL condition with constant μ . The convergence rate $(1 - \frac{\mu}{L})^T$ matches the classical linear rate for gradient descent on strongly convex functions.

4.4.2 Case 2: $\mu = L$

When $\mu = L$, the rate factor becomes $1 - \frac{\mu}{L} = 0$. This implies:

$$f(x_1) - f^* \leq 0 \implies f(x_1) = f^*.$$

The algorithm converges in one step. For example, consider $f(x) = \frac{L}{2}\|x\|^2$, which satisfies $\mu = L$. Gradient descent with $\eta = \frac{1}{L}$ gives:

$$x_{t+1} = x_t - \frac{1}{L} \cdot Lx_t = 0,$$

which is the global minimum.

4.5 Discussion

This proof demonstrates that gradient descent with fixed step size $\eta = \frac{1}{L}$ achieves linear convergence for L -smooth functions satisfying the PL condition. The rate depends critically on the ratio μ/L , with smaller μ/L leading to slower convergence. The analysis extends to non-convex functions satisfying the PL condition, generalizing the classical results for strongly convex objectives.